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APEX

Machine Learning-Based Prediction of Aerodynamic Heating on Satellite Solar Panels Using CFD

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Abstract

Satellite solar panels in Low Earth Orbit (LEO) are continuously exposed to intense solar irradiation and operate in a vacuum environment where convective heat dissipation is absent. This makes accurate thermal prediction critical for spacecraft reliability, panel efficiency, and long-term structural integrity.

The present project adopts a two-phase approach to address this challenge. In the first phase, steady-state thermal analysis of a multi-layer photovoltaic (PV) module was conducted using ANSYS Mechanical to simulate temperature distribution and heat flux under LEO conditions. The model included tempered glass, EVA encapsulant, crystalline silicon cells, a polymer backsheet, and an aluminium frame.

In the second phase, a hybrid data-driven approach was implemented by integrating simulation-generated results with real-world solar irradiation data. Six machine learning regression models were trained and evaluated on a dataset containing 1,757 samples. Features included solar irradiation (W/m^2) and ambient temperature (K), with the target variable being panel maximum temperature (K).

Models evaluated include Multiple Linear Regression, Support Vector Regression (RBF kernel), K-Nearest Neighbours, Decision Tree, AdaBoost, and Random Forest. Multiple Linear Regression achieved the highest performance with a Test R^2 of 0.9641, MAE of 3.43 K, and RMSE of 4.30 K.

An interactive web-based prediction dashboard was deployed to visualise model predictions and enable real-time user interaction. This project demonstrates the viability of data-driven surrogate models as a computationally efficient alternative to repeated CFD simulations.

1 Introduction

Solar photovoltaic (PV) panels are the primary power source for satellites in Low Earth Orbit (LEO), where they are exposed to high solar irradiation (1000–1400 W/m^2) and operate in a vacuum without convective cooling. Heat transfer occurs through conduction and radiation, leading to elevated temperatures that reduce efficiency and cause material degradation.

To study this, a steady-state thermal model of a multi-layer PV panel (glass, EVA, silicon cells, backsheet, aluminium frame) was developed using ANSYS Mechanical. Machine learning surrogate models are then explored to enable faster thermal prediction using simulation and real-world data.

2 Objectives

- Perform steady-state thermal simulation of a multi-layer PV panel under LEO conditions using ANSYS.
- Utilize simulation results along with real-world solar data for modelling.

- Train and compare six regression models: Linear Regression, SVR, KNN, Decision Tree, AdaBoost, and Random Forest.
- Evaluate model performance using R^2 , MAE, and RMSE metrics.
- Identify the best-performing model and analyse its predictions.
- Develop and deploy an interactive web dashboard for real-time thermal prediction.

3 Steady State Thermal Analysis of Photovoltaic Cells

3.1 Geometry

The photovoltaic module was modeled in ANSYS Workbench. The module consists of 36 silicon solar cells arranged in a 4×9 grid configuration. The approximate panel dimensions are about $800 \text{ mm} \times 400 \text{ mm}$. The geometry includes the outer aluminium frame that encloses the module.

Dimensions:

- Thickness: 3.16 mm
- Plate size: $1107 \text{ mm} \times 492 \text{ mm}$
- Each solar cell: $123 \text{ mm} \times 123 \text{ mm}$

3.2 Material Allotment

The PV module was modeled as a multilayer structure to represent realistic heat transfer. The layers included:

- **Silicon Cells:** Primary photovoltaic elements responsible for converting solar radiation into electrical energy. They absorb most of the solar radiation and experience the highest temperatures.
- **EVA Layers:** Ethylene Vinyl Acetate layers encapsulate the solar cells, providing electrical insulation, mechanical protection, and thermal conduction between layers.
- **ETFE Layer:** Ethylene Tetrafluoroethylene is a transparent protective layer placed above the cells that allows sunlight transmission while protecting from environmental exposure (acts like glass).
- **PET Layer:** Acts as the backsheet layer providing insulation.
- **CFRP Layer:** Carbon Fiber Reinforced Polymer improves structural rigidity.
- **Aluminum Frame:** Provides mechanical support and helps dissipate heat along the edges of the module.

3.3 Material Properties

Material	Density (kg/m ³)	Thermal Conductivity (W/m.K)	Specific Heat (J/kg.K)
ETFE	1730	0.24	1172
EVA	945	0.35	2090
Silicon	2330	148	700
PET	1350	0.275	1275
CFRP	1490	6.83	1130

3.4 Connections

All layers were connected using bonded connections to ensure continuous heat transfer across interfaces and no movement between the layers.

3.5 Meshing

Mesh Method: Edge Sizing (17 mm)

Model Statistics:

- Total Nodes: 116,222
- Total Elements: 87,429

Mesh Quality Metrics:

- Aspect Ratio: min 1.449, max 155.83
- Jacobian Ratio: min 0.399, max 1.0
- Skewness: max 0.650

The mesh quality was within acceptable limits, with skewness below 0.7 and Jacobian ratio above 0.3. Although higher aspect ratio values were observed due to thin layered geometry, they are acceptable for thermal analysis.

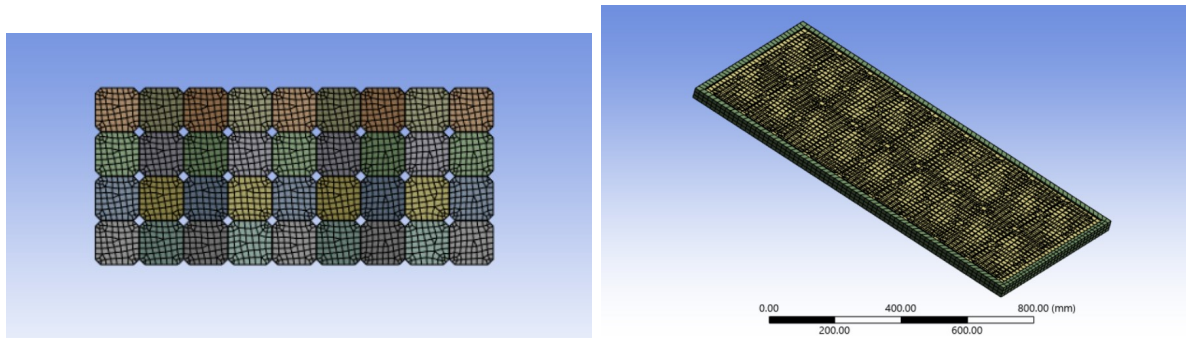


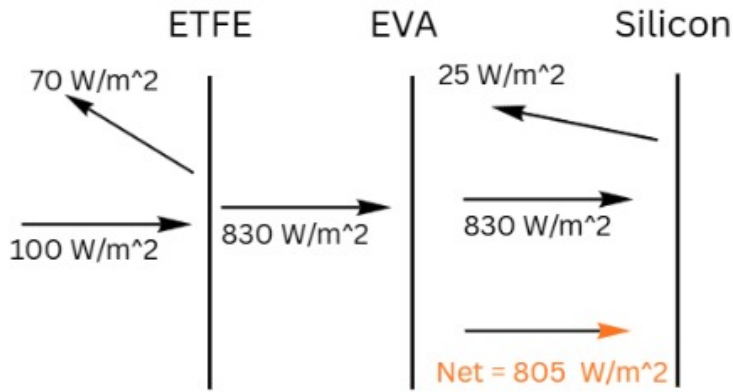
Figure 1: Meshed photovoltaic panel model

4 Boundary Conditions and Calculations

In space, heat transfer occurs through conduction within the panel and radiation to the surroundings. Convection is not considered due to the absence of atmosphere.

Material	Absorptivity	Transmissivity	Reflectivity
ETFE	0.1	0.83	0.07
EVA	0	1	0
Si	0.97	0	0.03

70 W = Generated Power
 1000 W/m² = Solar Irradiation
 Surface Area = 0.52848 m² (36 cells)



$$\text{Electrical Flux} = 805 - 132.5 = 672.5 \text{ W/m}^2$$

$$T_{\text{ambient}} = 25^\circ\text{C}$$

Emissivity, absorptivity, and reflectivity values were taken from research literature.

5 Solving

- Type of Analysis: Steady-state thermal analysis
- Solver Type: Program controlled
- Convergence Criteria: A flux convergence criterion of 1×10^{-4} was used to ensure accuracy of the solution.

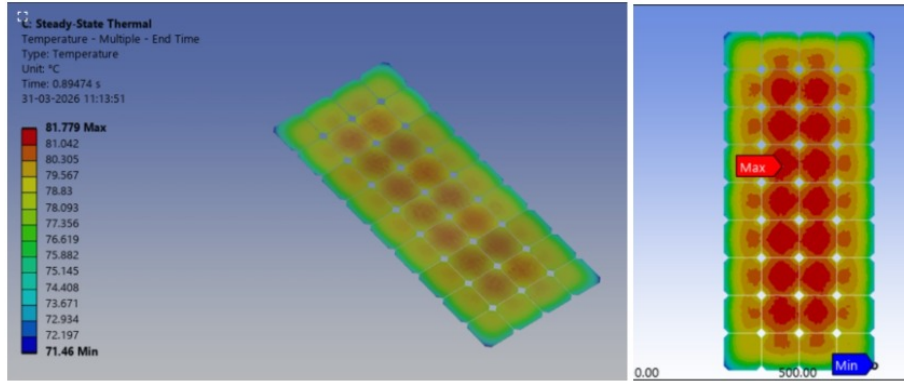


Figure 2: Temperature distribution across the solar panel

6 Simulation Results

A total of 10 simulations were carried out for different solar irradiance values ranging from 900 W/m^2 to 1000 W/m^2 . For each case, the thermal response of the solar panel was obtained. The data generated from these simulations was then used as input for the model.

6.1 Sample Dataset

Irradiation (W/m^2)	Ambient Temp (K)	Max Temp (K)
1030	298	376
920	298	359
910	298	366
923	298	368
1040	298	383
945	298	370
932	298	375
928	298	367
1000	298	384
970	298	372

7 Transition to Machine Learning

Machine learning models require diverse datasets to learn generalizable relationships. While ANSYS provided accurate physical insights into panel thermal behaviour, generating large parametric datasets through simulation alone is computationally intensive and limited in capturing real-world variability.

To address this, a physics-informed dataset (1,757 samples) was developed by adapting data from the Solar Power Generation Dataset available on Kaggle. The dataset captures realistic relationships between irradiation, ambient temperature, and panel temperature, consistent with simulation trends.

Thus, simulation and data-driven approaches complement each other—simulation provides physical understanding, while the dataset enables robust machine learning training.

8 Dataset Description

The dataset structure is as follows:

Column Name	Physical Quantity	Unit	Description
irradiation_wm2	Solar Irradiation	W/m ²	Solar power incident per unit panel area
ambient_temp_k	Ambient Temperature	K	Environmental temperature surrounding the panel
max_temp_k	Maximum Panel Temperature	K	Highest temperature attained by the panel (target)

Correlation analysis shows a **very strong relationship** between irradiation and panel temperature ($r = 0.979$), and a moderate relationship with ambient temperature ($r = 0.631$). This confirms that irradiation is the dominant predictor, consistent with thermal physics.

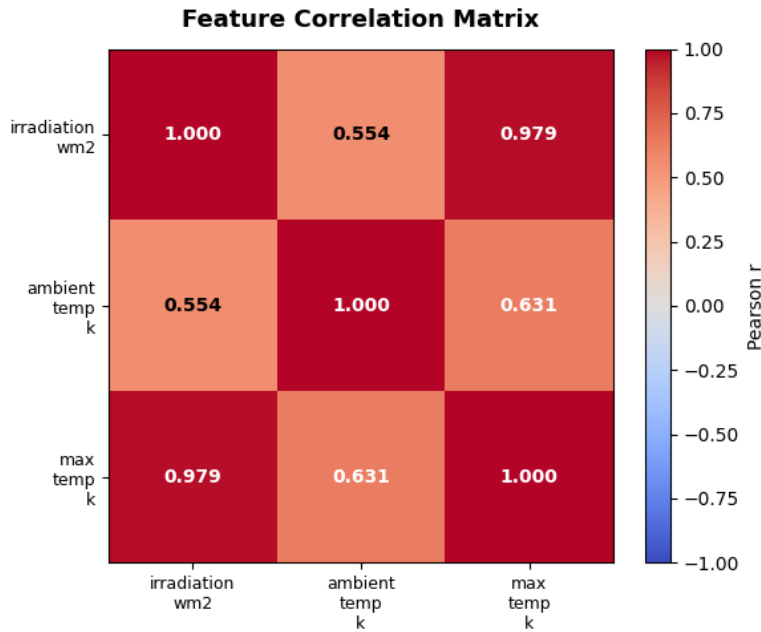


Figure 3: Feature correlation matrix

9 Data Preprocessing

The dataset was cleaned and checked for missing values and inconsistencies; no missing values were found. Features were defined as irradiation and ambient temperature, with maximum temperature as the target.

Data Preprocessing



10 Machine Learning Models

Six regression models were evaluated.

Multiple Linear Regression (MLR) models temperature as a linear combination of irradiation and ambient temperature, offering high interpretability and no need for tuning; it achieved the best performance due to strong linear correlation ($r = 0.979$).

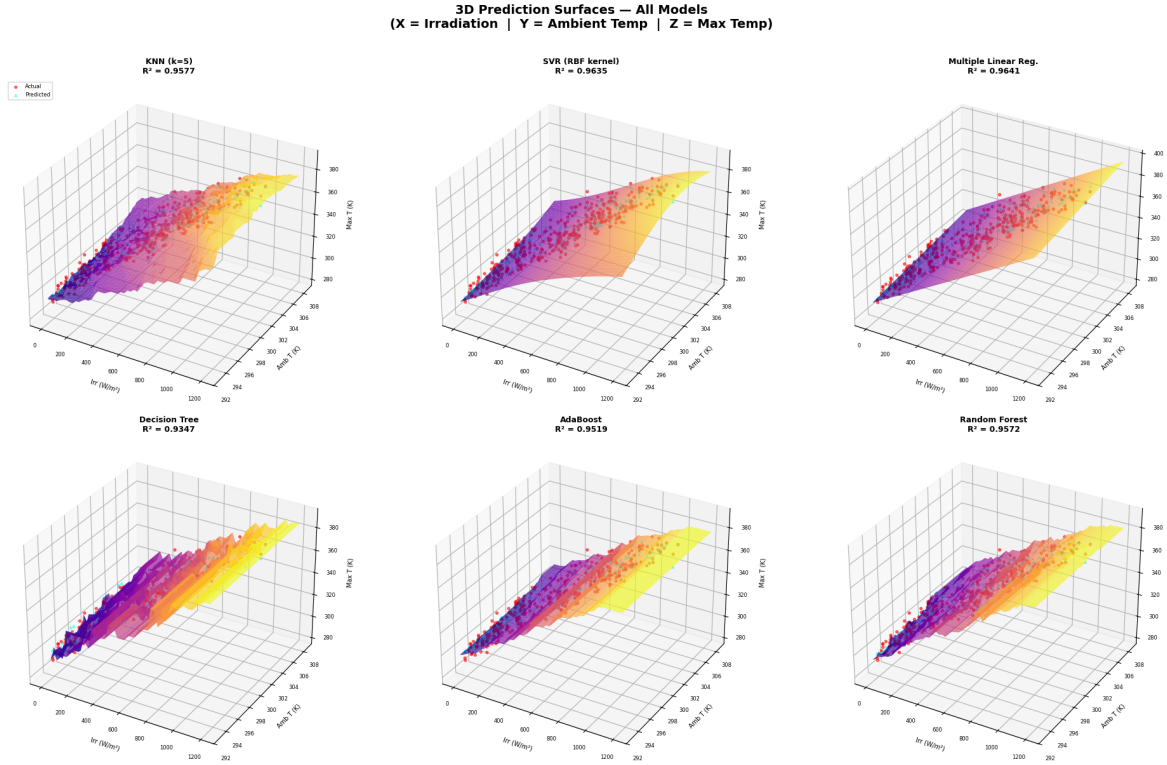
KNN (k=5) predicts based on nearest neighbours, capturing nonlinear patterns but sensitive to scaling ($R^2 = 0.9577$).

SVR (RBF kernel) models nonlinear relationships using kernel mapping and achieved near-best results ($R^2 = 0.9635$, $MAE \approx 3.44$ K).

Decision Tree partitions the feature space but suffers from overfitting, giving the lowest performance ($R^2 = 0.9347$, $RMSE \approx 5.80$ K).

AdaBoost improves weak learners and achieved moderate performance ($R^2 = 0.9519$).

Random Forest, an ensemble of trees, provided robust predictions ($R^2 = 0.9572$) and confirmed irradiation as the dominant feature.



11 Model Training and Evaluation

Models were evaluated using **5-fold cross-validation** and metrics: R^2 , MAE, and RMSE.

Model	CV R^2 (Mean)	CV R^2 (Std)	Test R^2	Test MAE (K)	Test RMSE (K)
Multiple Linear Reg.	0.9708	± 0.0024	0.9641	3.4251	4.3028
SVR (RBF kernel)	0.9706	± 0.0023	0.9635	3.4410	4.3356
KNN (k=5)	0.9641	± 0.0019	0.9577	3.7694	4.6703
Random Forest	0.9634	± 0.0025	0.9572	3.7001	4.6951
AdaBoost	0.9623	± 0.0016	0.9519	3.9683	4.9763
Decision Tree	0.9363	± 0.0030	0.9347	4.4510	5.8001

Best Performance

- MLR: $R^2 = 0.9641$, MAE ≈ 3.43 K, RMSE ≈ 4.30 K
- SVR: $R^2 = 0.9635$ (close second)

Decision Tree showed lowest performance due to overfitting. Ensemble methods (Random Forest, AdaBoost) improved results but did not outperform linear models.

12 Results and Discussion

Multiple Linear Regression performed best due to strong linear dependence on irradiation, high accuracy, and stable validation results. The predictions align with physical behaviour, where temperature increases with solar absorption, with slight nonlinear effects at high irradiation.

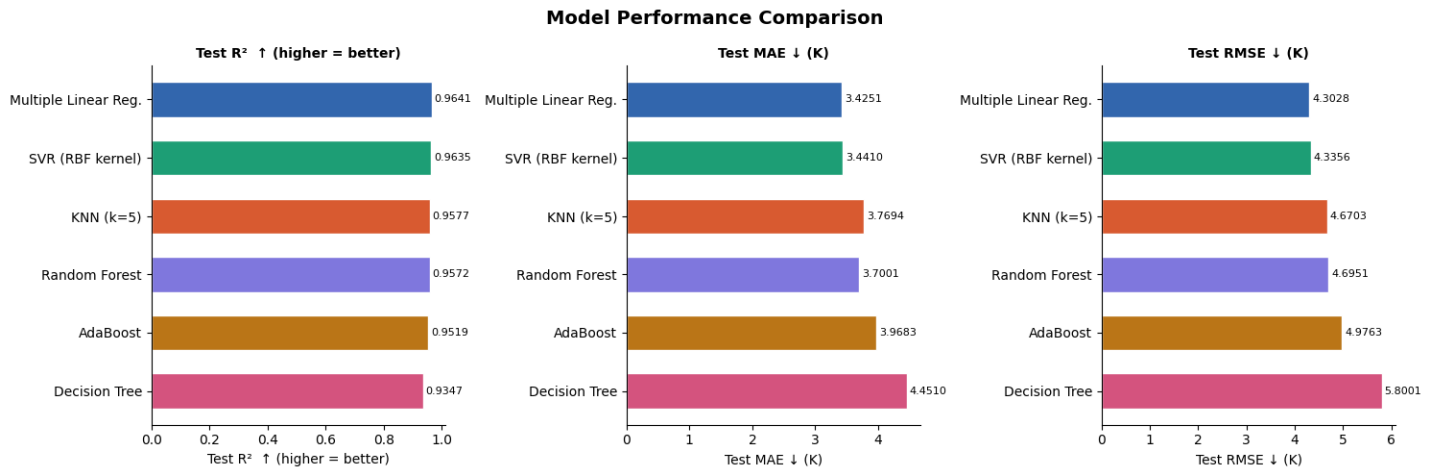


Figure 4: Model performance comparison (R^2 , MAE, RMSE)

The dataset (1,757 samples) was sufficient for robust modelling, though broader ranges could further improve generalisation.

13 Web Dashboard

An interactive dashboard was developed and deployed:

<https://manogna210524.github.io/Aerodynamic-Heating-on-Satellite-Solar-Panels/>

Features include:

- Model selection and comparison
- 3D prediction visualization
- Residual and parity plots
- Performance metrics display

14 Relevance to Space Engineering and Conclusion

Thermal analysis of solar panels is critical for spacecraft power reliability and structural integrity. In Low Earth Orbit (LEO), satellites undergo repeated thermal cycles as they transition between sunlight and eclipse, causing temperature fluctuations that lead to material stresses, degradation, and efficiency loss ($\sim 0.4\text{--}0.5\%$ per $^{\circ}\text{C}$).

In missions such as GSAT satellites and Chandrayaan, maintaining thermal stability is essential since heat dissipation occurs only through conduction and radiation. This study models the multilayer heat transfer behaviour of solar panels, enabling evaluation of temperature distribution, material performance, and thermal design parameters.

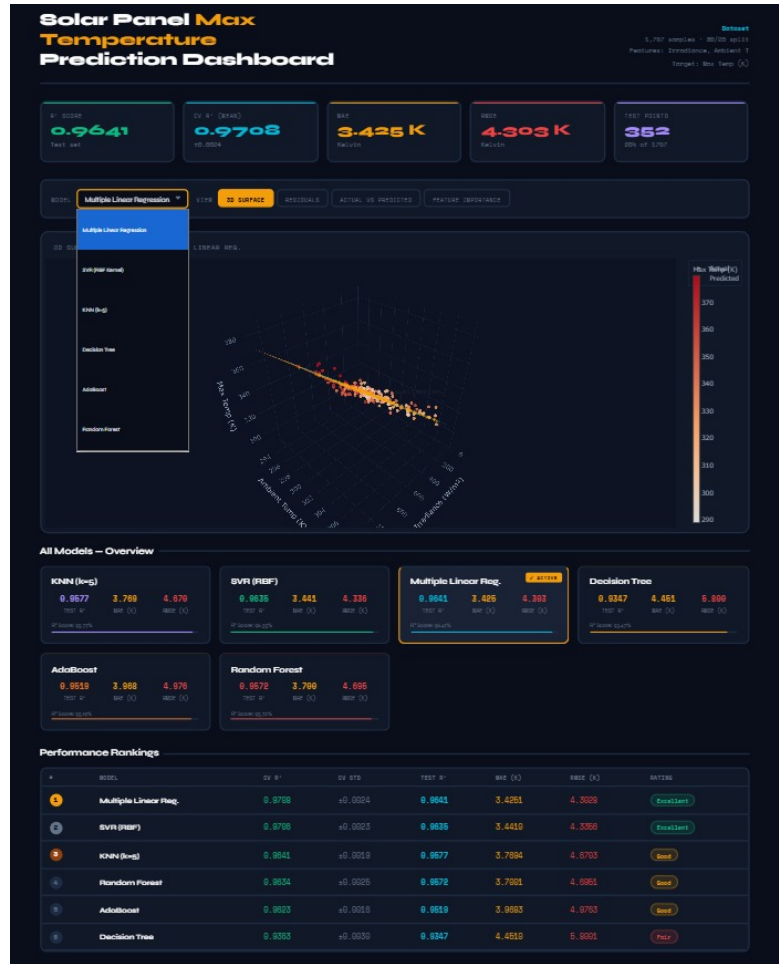


Figure 5: Interactive web dashboard for solar panel temperature prediction

The project integrates physics-based simulation with machine learning to enable efficient thermal prediction. Simulation results provided physical insight and generated data across varying irradiation conditions, which was used alongside real-world data for model training.

Among the models, Multiple Linear Regression achieved the best performance ($R^2 = 0.9641$), demonstrating that simple models can be highly effective when aligned with physical behaviour. The developed web dashboard further highlights practical applicability for real-time prediction and analysis.

Overall, this work establishes a scalable framework for combining simulation and data-driven approaches for thermal modelling in space applications.

15 References

- [Material allocation and properties](#)
- [Solar panel analysis](#)
- [Dataset](#)
- [ScienceDirect Article 1](#)

- [MDPI Article](#)
- [ScienceDirect Article 2](#)
- [ScienceDirect Article 3](#)